

SOLO SAATHI CIRCLE

TECHNICAL SPECIFICATION

Complete handoff reference for engineering

Algorithm · Architecture · Integrations · Database · Environment Setup

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Product: Solo Saathi Circle



1. Product Overview

Solo Saathi Circle matches solo Garba/Navratri attendees into evenly-sized, gender-balanced “Circles” for the night, led by a volunteer Captain. Two registration paths feed the same matching system:

- Live — walk-up registration at the venue, matched in real time
- Advance — pre-event registration, pooled and finalized in batches ahead of the event

Revenue: a flat per-attendee registration fee (₹199 standard / ₹249 on peak dates), paid directly by attendees. A separate organizer-facing “Certified” partnership model exists for venues (covered in business/marketing materials, not detailed here since it does not affect the core system).

2. System Architecture

2.1 Stack

- Frontend: React + Vite
- Backend: Netlify Functions (serverless, Node.js)
- Database: Netlify Blobs (key-value store — see Section 11 for the full schema)
- Deployment: Netlify (functions + static frontend from one project)

2.2 Deliverables — what each file is

Deliverable	What it is	Use it for
solosaathi-preview.jsx	Single-file React prototype, client-side simulation only, no real backend	The authoritative UX/visual reference — every screen, animation, and exact copy has been browser-tested
solosaathi-netlify-deployable.zip	Real backend (Netlify Functions + Blobs) + functional-core frontend	The actual codebase to build on. Backend logic is complete and unit-tested; frontend is functional but not visually polished

⚠ The preview file and the deployable project are NOT the same codebase. The preview has richer visuals; the deployable project has the real, working backend. Porting the preview's visual layer onto the deployable project's real backend is a core remaining task.

3. Core Matching Algorithm

3.1 Circle naming

Circle IDs follow {PREFIX}-{2-digit index}, where the prefix is determined by level and gender preference:

Level / Type	Prefix
Beginner	GARBA
Intermediate	TAAL
Advanced	RAAS
All-women (any level)	SAKHI

Other Group Name Options: TOLI , DHOL, DHAKLA, TARANG, MANDLI.



3.2 Gender balance rules

- GENDER_CAP = 10 — in a MIXED circle, no more than 10 people of one declared gender
- Unstated (“prefer not to say”) gender never counts toward the cap
- All-women circles bypass the gender cap entirely (there is no such thing as “too many women” in a women-only circle)
- ALL_WOMEN_MIN_FLOOR = 4 (preferred minimum before closing an all-women circle); ALL_WOMEN_ABSOLUTE_MIN = 3 (last-resort minimum) — this is a soft target, not a hard block

3.3 Group sizing

- SOFT_MAX_GROUP = 24 — no hard ceiling on circle size, but a new bucket starts once a bucket reaches this to avoid unbounded growth
- Live registration: continuous — each new registrant is checked against the current open bucket for that city/venue/level/gender-preference/date; shouldStartNewBucket() decides whether they join it or start a fresh one
- Advance registration: batch — registrants accumulate in a pending pool keyed the same way (plus event date) until the pool is finalized into one or more real circles

3.4 Bucket key structure

```
Live:    groupstate:{city}:{venue}:{level}:{genderPref}:{date}
Advance: pending:{city}:{venue}:{level}:{genderPref}:{eventDate}
```

This means a circle for Oct 15 can never accidentally merge with one for Oct 20 — date is always part of the key.

4. Registration Flows

4.1 Live registration

- Fields: ticket serial or photo, name, WhatsApp number, city, venue, gender, age band, all-women toggle, Captain opt-in, skill level
- Gated by isLiveRegistrationOpen(): must be on/after the live season start date (2026-10-13) AND within the daily window (6:30 PM – 1:30 AM IST, wraps past midnight)
- Geolocation venue-confirm: auto-detects GPS on the main CTA tap (1-2km radius), shows a confirm modal with override if it disagrees with the manually-picked venue
- Requires OTP-verified WhatsApp number before submission (see Section 5) — enforced server-side
- Price: ₹199 standard, ₹249 on configured peak dates (see shared/pricing.js)

4.2 Advance registration

- Same core fields, plus: mandatory ticket photo (no serial-only option — there is no physical-presence signal yet, so the photo is the only fraud check available), and a required Event Date picker
- Gated by isAdvanceRegistrationOpen(): open any time up until 2 hours before the specific event's 7:30 PM IST start — NOT a fixed 48-hour registration cutoff
- Feeds the pending pool, not instant matching

The two different ‘48-hour’ concepts — do not conflate them

⚠ (1) Registration eligibility stays open through day-before AND day-of — no hard cutoff. (2) The GROUP-FORMATION notification promise (“you’ll hear by 48 hours before the event, even if the group is small”) is a separate, softer commitment about finalization timing, not about when registration closes.



4.3 Finalizing a pending pool

- finalize-bucket.js converts a pending pool into a real circle: forms the roster, assigns the first Captain-opted-in entrant (or the first entrant) as Captain, writes a registration record for every real entrant, clears the pool

⚠ Production still needs a trigger: a Netlify Scheduled Function checking every open pool's event date and calling finalize-bucket.js when it crosses the 48-hour mark, OR an organic trigger when a pool crosses a healthy size (e.g. 12+). Currently this is a manually-callable, admin-authenticated endpoint only.

5. OTP Verification System

Added specifically to stop registration on a WhatsApp number the registrant doesn't control (preventing later refund/dispute abuse). Applies identically to Live and Advance.

5.1 Flow

- send-otp.js: generates a 6-digit code, stores it with a 10-minute expiry, sends via WhatsApp (primary) or SMS (fallback)
- verify-otp.js: checks the code; on success, marks the number “verified” for 30 minutes, checked by register.js / advance-register.js before accepting a registration

5.2 Hard block — by design, no override

⚠ After 5 incorrect attempts, the code is invalidated outright. There is no manual bypass. If someone can never enter the correct code (because it isn't their number), they simply cannot register. This was an explicit decision, not an oversight.

5.3 Rate limiting

Rule	Value
Max sends per number	3 per 15-minute window
Resend cooldown	30 seconds between sends
Code expiry	10 minutes
Max verify attempts	5, then hard-blocked until a new code is requested
Verified TTL	30 minutes to complete registration after verifying

5.4 WhatsApp setup

- Requires a Business Solution Provider (Interakt, AiSensy, or Gupshup are common in India) and Meta Business verification
- Submit an authentication-category template: “Your SoloSaathi verification code is {{1}}. Valid for 10 minutes. Do not share this code.” — authentication templates get faster, more standardized review than marketing templates
- Verified 2026 pricing: ~₹0.115 per message domestic, ~₹2.30 international. No DLT registration needed for WhatsApp (unlike SMS)
- SMS fallback (if in scope): needs a DLT-registered sender + template via a provider like MSG91 or Kaleyra — budget 1-2 weeks for DLT approval, a real lead time separate from WhatsApp setup

6. AI Ticket Verification

⚠ This MUST run server-side. The original prototype called Anthropic's API directly from the browser — that only worked inside the special claude.ai artifact runtime, which auto-injects credentials. A plain client-side fetch() to api.anthropic.com from an ordinary deployed site fails outright. verify-ticket.js exists specifically to keep the API key secret on the server.



- Accepts a base64 image, calls Anthropic's vision API (model claude-sonnet-4-6) with a structured-extraction prompt
- Returns: city, venue, venue_english, pass_id, looks_like_valid_ticket, plus a ticketVerifiedToken the frontend passes along to advance-register.js as proof the check ran
- Mandatory for Advance registration; for Live, a ticket serial number OR a photo is acceptable (physical presence + geolocation already provide some fraud signal)
- Venue-mismatch check: compares the AI-extracted venue against the manually-selected venue and warns the user if they don't match — present in both Live and Advance flows

7. Chat & Circle Management

7.1 Core chat features

- Free text plus quick-tap chips (not chip-restricted)
- Profanity filter: masks flagged words with asterisks, shows a gentle nudge
- Separate spam/ad-pattern detector (URLs, phone numbers, promo keywords): blocks the message outright and triggers a 60-second cooldown on that user's input
- Report button on any message — flags to a volunteer, does not auto-remove content

7.2 Grow / Lock / Leave / Transfer — real shared state

Backed by a `circlestate:{circleId}` record in Blobs — this is the one piece that genuinely could not be real in a client-only prototype, since there was no second real user to keep in sync.

- Grow: any member can open 1-20 more spots, unlocks the circle if it was locked
- Lock: any member can close the circle to new members — resolved once per chat session
- Leave: any member can leave, freeing their slot for someone else
- Transfer Captain: the current Captain can hand the role to anyone in the roster; includes a Cancel option to back out without transferring

7.3 Switch Circle

- Capped at 3 switches per night
- Enabled only after 5 minutes since joining the current circle, then a 20-minute cooldown between switches

7.4 Auto-close

Chat and the Circle itself fully disappear (not read-only) at 1:00 AM IST. A prior-notice system message warns of this when chat opens. The exact locked closing message:

“That's a wrap, Saathi! 🌊 Raas over, circle closed — feet tired, heart fuller. 🧡 Same ground, same magic, tomorrow?”

⚠️ *This exact wording should not be altered without their sign-off.*

8. Organizer Dashboard

- Login: Venue ID + Password only (e.g. AH-GMDC / gmhc2026) — deliberately does NOT expose a browsable list of other venues

⚠️ *Current ID/password derivation is a deterministic formula from the venue name — fine for early testing, not for production. Replace with a real per-venue credential store before onboarding actual organizers.*



- Returns AGGREGATE stats only: total registrations (Advance vs Live split), level breakdown, gender balance, circles formed + average size, all-women circle count, show-up rate

⚠ Explicitly excludes money, names, and phone numbers — enforced server-side in `organizer-stats.js`, not just hidden in the UI.

9. Find My Circle

- Looks up every registration (both Live and Advance, across every night) for a given WhatsApp number, most recent first
- Today's / currently-live entries get full Beacon + Chat access; past entries show read-only details plus their own QR/token — Beacon and Chat are correctly withheld once a night has passed

10. Database Schema — Netlify Blobs Key Reference

Key pattern	Holds
<code>registration:{id}</code>	One attendee's full registration record (Live or Advance)
<code>bymobile:{whatsapp}:{date}</code>	Index from a WhatsApp number + date to a registration id, for Find My Circle
<code>pending:{city}:{venue}:{level}:{pref}:{eventDate}</code>	Array of not-yet-finalized Advance registrants for one bucket
<code>groupstate:{city}:{venue}:{level}:{pref}:{date}</code>	Running Live-matching state for one bucket (current group index, gender counts)
<code>circlestate:{circleId}</code>	Real-time chat state: locked flag, open slots, Captain name, action log
<code>showups:{city}:{venue}</code>	Array of post-event yes/no show-up responses, read by the Organizer Dashboard
<code>otp:{whatsapp}</code>	Active OTP code, expiry, attempt count
<code>otp-rl:{whatsapp}</code>	OTP send rate-limit window and count
<code>verified:{whatsapp}</code>	Timestamp of last successful OTP verification (checked against the 30-min TTL)
<code>venues</code>	The full city → venue directory (name, lat/lng), editable via the admin panel

11. Third-Party Integrations Required (“Connections”)

Integration	Purpose	Status
WhatsApp Business API (via a BSP)	OTP delivery + group-formed / reminder notifications	Not connected — needs a real BSP account + approved templates
SMS gateway (DLT-registered)	OTP fallback if WhatsApp delivery fails	Not connected — optional, agreed as in-scope
Anthropic API	Server-side AI ticket photo verification	Not connected — needs a real API key
Payment gateway (Razorpay / Cashfree)	Actually charging the ₹199/₹249 registration fee	Not built at all yet — current functions do not charge money
Netlify Blobs	Primary database	Already wired and working



Netlify Scheduled Functions	Auto-triggering finalize-bucket.js at the 48h mark	Not built — endpoint exists, nothing calls it on a schedule yet
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12. Environment Variables

Variable	Purpose
ADMIN_SECRET	Protects the admin venue panel and finalize-bucket.js
WHATSAPP_API_KEY / WHATSAPP_API_URL	Your BSP's credentials and send endpoint
SMS_API_KEY / SMS_API_URL / SMS_DLT_TEMPLATE_ID	SMS fallback provider credentials
ANTHROPIC_API_KEY	Server-side AI ticket check
OTP_DEV_MODE	Set to “true” ONLY for local testing — echoes OTP codes when no real provider is configured. NEVER set in production

13. Pricing Rules

- BASE_PRICE = ₹199, PEAK_PRICE = ₹249
- Peak dates are configured in shared/pricing.js — currently placeholder dates, must be set to the real Navratri peak nights (typically weekends and the last day or two) before launch

14. What's Built vs What's Pending

Area	Status
Matching algorithm (gender cap, all-women, sizing)	Built and unit-tested
Live + Advance registration logic	Built and unit-tested
OTP verification (hard block, rate limits)	Built and unit-tested
IST time-gating (season start, daily window, chat close)	Built and unit-tested — a real timezone bug was caught and fixed during development, worth re-testing if this code is touched
Organizer Dashboard (aggregate stats)	Built and unit-tested
Circle actions (grow/lock/leave/transfer/showup)	Built
Real WhatsApp/SMS sending	Not built — clearly marked TODOs with the exact integration shape expected
Payment gateway	Not built at all
Scheduled auto-finalization	Not built — endpoint ready, no scheduler wired yet
Full visual/UX polish on the real frontend	Not ported — use the preview file as the reference
End-to-end testing against a live Netlify deployment	Not possible from the sandbox this was built in — only unit-level logic testing was done



15. Brand & Design Reference

- solosaathi-preview.jsx is the authoritative source for exact colors, copy, animations, and interaction details — refer to it whenever a visual question comes up that this document doesn't answer
- Primary palette: To be considered as per the Logo Shared.
- Logo: white-background version for light contexts, white text version for dark contexts — both delivered separately as PNG assets

16. Legal & Business Context (for reference, not engineering)

- Terms & Conditions: movie-ticket-style non-refund/non-transfer policy, 18+ only, drafted but not yet lawyer-reviewed
- Collaborator NDA + IP Assignment and Onboarding Process documents exist separately for hiring freelancers/volunteers

These don't affect how the system is built, but the developer should know the non-refund policy exists since it's referenced in the UI copy at multiple points in the registration flow.